

**GENFT 302: Applied Mathematics /
Apply Fundamental Algebra and
Trigonometry**

Date: 12 June 2024

Period: 08.30 am – 11.30 am



END OF TERM III EXAMINATIONS QUESTION PAPER

SECTOR: ICT & MULTIMEDIA

TRADE: Computer Systems and Architecture (CSA), Multimedia Production (MMP), Software Development (SFD), Networking and Internet Technologies (NIT).

TVET LEVEL: 3

MODULE TITLE: Applied Mathematics/ Apply Fundamental Algebra and Trigonometry

MARKS:

100

CAMIS

...../60

DURATION: 3 HOURS

INSTRUCTIONS TO CANDIDATES:

This Exam paper is composed of Three Sections (A, B). Follow the instructions given below, and answer the indicated questions for a total of 100 marks

Section A: Fifteen (15) questions, all Compulsory 55 marks

Section B: Among the five (5) questions, attempt any three (3) 45 marks

SECTION A: Attempt All questions**[55 marks]**

1. Given the parametric equation $f(x) = \frac{mx}{2m+3}$, state the condition for which the parametric linear equation with parameter m has the solution

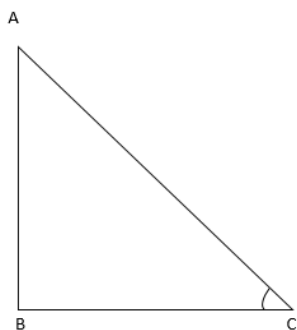
[2 marks]

2. Define the following terms

[3 marks]

- a. Light angle
- b. Obtuse angle
- c. Reflex angle

3. Consider the given triangle and determine the $\cos \hat{c}$, $\sin \hat{c}$ and $\tan \hat{c}$.

[3 marks]

4. If matrix $A = \begin{pmatrix} 7 & 1 & -3 \\ 2 & 4 & 6 \\ 0 & z & -1 \end{pmatrix}$ and $B = \begin{pmatrix} 7 & 1 & -3 \\ 2 & x & 6 \\ 0 & 3 & y \end{pmatrix}$ are equal, find the value

of x , y and z

[3 marks]

5. Define the following terms:

- a. Upper triangular matrixes
- b. Diagonal matrix
- c. Scalar matrix
- d. Identity matrix
- e. Lower triangular matrix

[1 mark]**[1 mark]****[1 mark]****[1 mark]****[1 mark]**

6. Find the determinant of this given matrix $A = \begin{pmatrix} 17 & 1 & -23 \\ 0 & 0 & 6 \\ 0 & 0 & -1 \end{pmatrix}$

Justify the answer

[3 marks]

7. Without using a calculator, find the value of ϕ if $\begin{cases} \sin \phi = \frac{\sqrt{3}}{2} \\ \cos \phi = -\frac{1}{2} \end{cases}$ [2 marks]

8. Considering the following matrices, which ones are square matrices. [3 marks]

$$A = \begin{pmatrix} 3 \\ 8 \\ 1 \end{pmatrix}, B = \begin{pmatrix} 0 & 5 \\ 2 & 1 \\ 5 & 4 \end{pmatrix}, C = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & k \end{pmatrix}, D = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, E = \begin{pmatrix} 1 & p \\ 6 & 2 \end{pmatrix}, F = (1 \ 3)$$

9. Solving the following equation $\begin{cases} y = x^2 \\ y = x + 6 \end{cases}$ [5 marks]

10. Compute the determinant of matrix $A = \begin{pmatrix} 2 & 5 \\ -5 & 0 \end{pmatrix}$ [2 marks]

11. Solve the simultaneous linear equation $\begin{cases} 3x - y = 6 \\ 2x + 4y = 4 \end{cases}$ [5 marks]

12. Using the following simultaneous linear equation: $\begin{cases} 3x - y = 0 \\ x - 2(x - y) = 1 \end{cases}$

a. Write this system of equation in the $AX = B$ form [2 marks]

b. Find the determinant of matrix A [2 marks]

13. Solve the following equation and inequality [5 marks]

a. $x^2 - 4x + 3 = 0$

b. $6x^2 - 5x + 1 \leq 0$

14. Solve the equation $\cos 2x + \cos 4x = 0$ [5 marks]

15. Given that $f(x) = x^2 - 4x + 3$

a. Write the name of the given equation [1 marks]

b. Find x -intercepts; [4 marks]

SECTION B: Attempt any Three (3) questions**[45 marks]**

16. a. Simplify $\left(\frac{1}{\tan x} + \frac{1}{\cot x}\right) \sin x \cos x$ **[7marks]**
- b. Verify if $\frac{\cos^2 x}{1 + \tan^2 x} - \frac{\sin^2 x}{1 + \tan^2 x} = 1 - 2\sin^2 x$ **[8 marks]**
17. a. The sum of two numbers is 25. One of the numbers exceeds the other by 9. Find the numbers. **[7 marks]**
- b. If $C = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 2 & 1 \\ 0 & 2 & 3 \end{pmatrix}$ and $D = \begin{pmatrix} 1 & 3 & -1 \\ -1 & 2 & 2 \\ 0 & 1 & -1 \end{pmatrix}$, what do you conclude about $|C \times D|$ and $|C| \times |D|$ **[8 marks]**
18. A triangular shape site has the measurements of 8m, 14m and 19m respectively at its sides. Determine the angles of that site. **[15 marks]**
19. Solve graphically the following equation and inequality
- a. $2x + 5y \geq 6$ and shade the region which contains the solution **[7 marks]**
- b. $\begin{cases} 3x - 4y = 1 \\ x - 3y = 2 \end{cases}$ **[8 marks]**
20. Find the exact values of the five trigonometric functions of an angle ϕ , in standard position given that $\cos \phi = \frac{3}{5}$ and the angle ϕ is in first quadrant **[15 marks]**

END OF ASSESSMENT!